

Member 2 Analytics Implementation Summary

Based on Group_033.pdf — Wijesingha S M — Park Management and Patrol Analytics Reports

Generated: 9 October 2026

Executive summary

Member 2 functionality has been implemented as a live MongoDB-backed analytics workspace. The workflow supports report selection, criteria configuration, analytics generation, saved reports, comparison, zone drill-down, CSV/PDF export, audit records, and a professional responsive dashboard. The application is configured for the requested no-login demonstration mode.

Overall assessment: Member 2 is substantially implemented. Final live database end-to-end verification and production authentication remain outside the current no-login scope.

Report types

- Patrol Coverage Report — patrol hours, required hours, coverage percentage, incident density, zone status, trends, recommendations and partial-data warnings.
- Incident Summary Report — incident totals, category distribution, zone breakdown, weekly trends, hotspots and incident-focused visual sections.
- Terrain Trend Report — weekly trends, terrain hotspots, density map, baseline comparison and risk-oriented recommendations.

Implemented backend functionality

- Live retrieval of incident and patrol records from MongoDB. No mock or seed analytics records are used.
- Date, terrain-zone, category and baseline validation. Date-only end dates include the complete final day.
- Nearest-sector assignment for incident density, zone statistics and drill-down consistency.
- Patrol coverage calculations against configured required sector hours.
- Weekly incident trend aggregation scoped to selected terrain sectors.
- Partial report status when selected sectors have no patrol records, with explanatory reasons.
- Data-driven recommendations for low coverage and elevated incident density.
- Sequential report IDs, saved report persistence and audit logging for generation and export.

Analytics API

POST /api/analytics/generate

GET /api/analytics/reports

GET /api/analytics/reports/:id

GET /api/analytics/reports/:id/export?format=csv|pdf

GET /api/analytics/zones/:code/records

Frontend screens and UX

- Report type selection, criteria form, results dashboard and saved reports pages.
- KPI cards, trend chart, coverage bars, density map, zone table and comparison table.
- Zone drill-down drawer with incident and patrol tabs, filtering, loading, error and empty states.
- Responsive sidebar/topbar, shared cards, consistent buttons, mobile layouts and export feedback.

Document requirement comparison

Report type selection: Complete

Date, terrain and category filters: Complete

Incident and patrol aggregation: Complete

Coverage, density and trends: Complete for the implemented analytics model

Summary and per-zone results: Complete

CSV/PDF export: Complete

Zone drill-down: Complete

Baseline comparison: Implemented with zone-level comparison

Partial patrol data: Implemented at basic sector level

No-data and invalid criteria flows: Complete

Database and export exception states: Implemented; live failure testing still required

Saved reports and audit records: Complete

Authorized manager login: Intentionally omitted per no-login instruction

Remaining work

- Run full end-to-end tests against a running MongoDB instance using real incident and patrol records.
- Improve partial-data detection to identify gaps within part of a reporting period, not only sectors with zero patrol records.
- Further separate Incident Summary and Terrain Trend layouts from shared patrol-coverage sections.
- Add richer category-by-period trend comparisons and increasing/stable/decreasing labels for Terrain Trend.
- Restore authentication and ownership filtering before production deployment.

Key implementation files

Backend: backend/services/analyticsService.js, backend/controllers/analyticsController.js, backend/services/exportService.js, backend/models/GeneratedReport.js, backend/routes/analytics.js.

Frontend: frontend/web/src/pages/analytics, frontend/web/src/components/analytics, frontend/web/src/services/analyticsApi.js.

Validation status

- Frontend production build passed during the latest completed validation.
- Backend JavaScript syntax checks passed during previous validation runs.
- git diff --check passed with only normal Windows line-ending warnings.
- This is an implementation summary; it does not contain live database report values.